Technical Supplement to
Baseline-Conditioned Source Attribution for DLMP Settlement with Price-Anticipating Load Response

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Abstract

Under strong convexity and effective-space linear independence of the active constraint gradients, the smooth penalty minimizers and network multiplier estimates recover the centralized hard-constrained LinDistFlow optimum and its KKT multipliers in the iterated limit: first $\varepsilon \downarrow 0$ at fixed penalty scale, then $\kappa \to \infty$. This supplement proves the corresponding voltage- and congestion-price limits. Eliminating fixed load coordinates gives the nondegenerate decision space on which the regularity condition is imposed. A separate analysis derives the source-attribution feedback Jacobian, continuous path moments, and curvature bounds used to assess strong monotonicity of the responsibility-settlement VI. The OPF limiting result and the QNE uniqueness criterion concern different solution concepts and retain their respective assumptions.

Sections S1–S4 establish and interpret the centralized OPF limiting result in Theorem S1. Section S5 provides the technical derivations for the main manuscript's QNE monotonicity analysis.

S1 Effective-space hard-constrained problem

S1.1 Elimination of rigid coordinates

Let the flexible coordinates be collected in $\mathbf{y} \in \mathcal{X} \subset \mathbb{R}^d$, where

$$\mathcal{X} := \prod_{h=1}^{d} [\underline{y}_h, \overline{y}_h], \qquad \underline{y}_h < \overline{y}_h. \tag{S1}$$

Here $(\mathbf{y}, \mathcal{X}, d)$ corresponds to the manuscript's $(\mathbf{y}_a, \mathcal{Y}_a, n_a)$. The embedding matrix G_a and fixed-load vector $\mathbf{p}^{\text{fix}}$ retain their manuscript definitions. Baseline-fixed loads enter the full non-root-bus power vector through

$$\mathbf{p}(\mathbf{y}) = \mathbf{p}^{\text{fix}} + G_a \mathbf{y}. \tag{S2}$$

Each retained interval has positive width, so the constraint qualification below is imposed directly on the free coordinates.

Define the reduced centralized objective

$$F(\mathbf{y}) := \mathcal{S}(\mathbf{p}(\mathbf{y})), \tag{S3}$$

where $\mathcal{S}$ is the centralized welfare objective in the manuscript. The scalar F in this supplement is distinct from the manuscript's vector VI operators F_p and F_y. Fixed-load utility constants do not affect the minimizers or derivatives. Under the main-study assumptions,

$$\nabla^2 F(\mathbf{y}) = D_{\text{util},a} + \frac{2\pi^E}{v_0} G_a^{\mathsf{T}} H^L G_a \succ 0, \tag{S4}$$

because $D_{\text{util},a} = \text{diag}(\beta_{u,h}) \succ 0$, $\pi^E \geq 0$, and $H^L \succeq 0$. Hence F is continuously differentiable and strongly convex on $\mathcal{X}$.

For the main-study lower-voltage and forward-flow limits, set

$$g_m^V(\mathbf{y}) := \underline{v} - v_m(\mathbf{p}(\mathbf{y})), \tag{S5}$$

$$g_\ell^C(\mathbf{y}) := f_\ell(\mathbf{p}(\mathbf{y})) - \overline{f}_\ell. \tag{S6}$$

Stack all network inequalities as $g_i(\mathbf{y}) \leq 0$, $i = 1, \ldots, r$. These functions are affine. Let

$$G := [\nabla g_1 \ \cdots \ \nabla g_r] \in \mathbb{R}^{d \times r}. \tag{S7}$$

The hard-constrained effective-space problem is

$$\min_{\mathbf{y} \in \mathcal{X}} F(\mathbf{y}) \quad \text{subject to} \quad g(\mathbf{y}) \leq 0. \tag{S8}$$

S1.2 KKT multipliers and scarcity-price mapping

Assume that problem (S8) is feasible, and denote its unique minimizer by $\mathbf{y}^\star$. The uniqueness follows from Eq. (S4). Let

$$\mathcal{A}_N^\star := \{i : g_i(\mathbf{y}^\star) = 0\} \tag{S9}$$

be the active network-constraint set. Let $B^\star$ collect one outward normal for every active bound of the nondegenerate box $\mathcal{X}$ at $\mathbf{y}^\star$. We impose the effective-space LICQ

$$[\, B^\star \quad G_{\mathcal{A}_N^\star} \,] \quad \text{has full column rank,} \tag{S10}$$

where $G_{\mathcal{A}_N^\star}$ contains the columns of G indexed by $\mathcal{A}_N^\star$. Unlike a full-space LICQ imposed on two inequalities for a zero-width interval, Eq. (S10) is compatible with rigid loads because those coordinates have already been eliminated.

The hard KKT system is

$$0 \in \nabla F(\mathbf{y}^\star) + G\boldsymbol{\omega}^{\text{h}} + N_{\mathcal{X}}(\mathbf{y}^\star), \tag{S11}$$

$$g(\mathbf{y}^\star) \leq 0, \qquad \boldsymbol{\omega}^{\text{h}} \geq 0, \tag{S12}$$

$$\omega_i^{\text{h}} g_i(\mathbf{y}^\star) = 0, \qquad i = 1, \ldots, r. \tag{S13}$$

Here $N_{\mathcal{X}}$ is the convex-analysis normal cone. Condition (S10) makes the complete KKT multiplier vector, and hence the network multiplier vector $\boldsymbol{\omega}^{\text{h}}$, unique.

Partition $\boldsymbol{\omega}^{\text{h}}$ into the low-voltage multipliers $\boldsymbol{\lambda}^{\text{h}}$ and the forward-flow multipliers $\boldsymbol{\nu}^{\text{h}}$. Since

$$\frac{\partial g_m^V}{\partial p_k} = A_{mk}^V, \qquad \frac{\partial g_\ell^C}{\partial p_k} = A_{\ell k}^C, \tag{S14}$$

the corresponding full-power marginal terms are

$$\pi_k^{V,\star} = \sum_m A_{mk}^V \lambda_m^{\text{h}}, \qquad \pi_k^{C,\star} = \sum_\ell A_{\ell k}^C \nu_\ell^{\text{h}}. \tag{S15}$$

This establishes the sign and sensitivity structure used in the hard DLMP.

S2 Quadratic and smooth outer-penalty problems

S2.1 Positive-part smoothing

For $z \in \mathbb{R}$, let

$$q(z) := \frac{1}{2}[z]_+^2, \qquad s_\varepsilon(z) := \varepsilon \log(1 + e^{z/\varepsilon}), \qquad \varphi_\varepsilon(z) := \int_{-\infty}^{z} s_\varepsilon(t)\, dt. \tag{S16}$$

The function q is continuously differentiable with $q'(z) = [z]_+$, whereas $\varphi_\varepsilon \in C^\infty$ and $\varphi'_\varepsilon = s_\varepsilon$. The standard Softplus bounds are

$$0 \le s_\varepsilon(z) - [z]_+ \le \varepsilon \log 2, \qquad z \in \mathbb{R}. \tag{S17}$$

Moreover,

$$0 \le \varphi_\varepsilon(z) - q(z) \le \frac{\pi^2}{6}\varepsilon^2, \qquad z \in \mathbb{R}. \tag{S18}$$

Indeed, the derivative of the difference in Eq. (S18) is nonnegative, and its total increase over $\mathbb{R}$ is

$$2\varepsilon^2 \int_0^\infty \log(1 + e^{-u})\, du = \frac{\pi^2}{6}\varepsilon^2. \tag{S19}$$

These bounds also show why smoothing errors in the multiplier coefficients are scaled by the penalty parameter.

S2.2 Penalized minimizers and multiplier estimates

For this proof, parameterize the manuscript's penalties at fixed positive ratios as $\kappa_V = \rho_V \kappa$ and $\kappa_C = \rho_C \kappa$. Each constraint weight ρ_i is either ρ_V or ρ_C. The quadratic outer-penalty problem is

$$\mathbf{y}^{(\kappa)} := \operatorname*{argmin}_{\mathbf{y}\in\mathcal{X}} \left\{ F(\mathbf{y}) + \kappa \sum_{i=1}^{r} \rho_i q(g_i(\mathbf{y})) \right\}. \tag{S20}$$

Its multiplier estimates are

$$\omega_i^{(\kappa)} := \kappa \rho_i [g_i(\mathbf{y}^{(\kappa)})]_+. \tag{S21}$$

The smooth outer-penalty problem is

$$\mathbf{y}^{(\kappa,\varepsilon)} := \operatorname*{argmin}_{\mathbf{y}\in\mathcal{X}} \left\{ F(\mathbf{y}) + \kappa \sum_{i=1}^{r} \rho_i \varphi_\varepsilon(g_i(\mathbf{y})) \right\}, \tag{S22}$$

with smooth multiplier estimates

$$\omega_i^{(\kappa,\varepsilon)} := \kappa \rho_i s_\varepsilon(g_i(\mathbf{y}^{(\kappa,\varepsilon)})). \tag{S23}$$

Strong convexity of F, convexity of q and φ_ε, and affinity of g_i make both penalized minimizers unique. Their stationarity systems are

$$0 \in \nabla F(\mathbf{y}^{(\kappa)}) + G\boldsymbol{\omega}^{(\kappa)} + N_{\mathcal{X}}(\mathbf{y}^{(\kappa)}), \tag{S24}$$
$$0 \in \nabla F(\mathbf{y}^{(\kappa,\varepsilon)}) + G\boldsymbol{\omega}^{(\kappa,\varepsilon)} + N_{\mathcal{X}}(\mathbf{y}^{(\kappa,\varepsilon)}). \tag{S25}$$

For voltage and congestion constraints, Eq. (S23) is exactly

$$\mu_m^{(\kappa,\varepsilon)} = \kappa_V s_\varepsilon\Big(\underline{v} - v_m(\mathbf{p}(\mathbf{y}^{(\kappa,\varepsilon)}))\Big), \quad \nu_\ell^{(\kappa,\varepsilon)} = \kappa_C s_\varepsilon\Big(f_\ell(\mathbf{p}(\mathbf{y}^{(\kappa,\varepsilon)})) - \overline{f}_\ell\Big). \tag{S26}$$

S3 Convergence results

S3.1 Removal of the smoothing error

Proposition S1 (Fixed-κ smoothing limit). *For every fixed $\kappa > 0$,*

$$\mathbf{y}^{(\kappa,\varepsilon)} \longrightarrow \mathbf{y}^{(\kappa)}, \qquad \boldsymbol{\omega}^{(\kappa,\varepsilon)} \longrightarrow \boldsymbol{\omega}^{(\kappa)} \quad \text{as } \varepsilon \downarrow 0. \tag{S27}$$

Proof. By Eq. (S18), the two objectives in Eqs. (S20) and (S22) satisfy the uniform estimate

$$0 \le P_{\kappa,\varepsilon}(\mathbf{y}) - P_\kappa(\mathbf{y}) \le \frac{\pi^2}{6}\kappa\varepsilon^2 \sum_{i=1}^{r} \rho_i, \qquad \mathbf{y} \in \mathcal{X}. \tag{S28}$$

Thus $P_{\kappa,\varepsilon} \to P_\kappa$ uniformly on the compact set $\mathcal{X}$. Every accumulation point of $\mathbf{y}^{(\kappa,\varepsilon)}$ minimizes P_κ. Since this minimizer is unique, the entire sequence converges to $\mathbf{y}^{(\kappa)}$.

For each i, Eq. (S17) and the 1-Lipschitz property of $[\cdot]_+$ give

$$\begin{aligned} \left|\omega_i^{(\kappa,\varepsilon)} - \omega_i^{(\kappa)}\right| &\le \kappa\rho_i\varepsilon \log 2 \\ &\quad + \kappa\rho_i \left| g_i(\mathbf{y}^{(\kappa,\varepsilon)}) - g_i(\mathbf{y}^{(\kappa)})\right|. \end{aligned} \tag{S29}$$

Affinity of g_i and convergence of the minimizers make the right-hand side tend to zero. □

S3.2 Quadratic-penalty limit

Proposition S2 (Quadratic outer-penalty consistency)**.** *Under the assumptions in Section S1, including the effective-space LICQ in Eq. (S10),*

$$\mathbf{y}^{(\kappa)} \longrightarrow \mathbf{y}^\star, \qquad \boldsymbol{\omega}^{(\kappa)} \longrightarrow \boldsymbol{\omega}^{\mathrm{h}} \quad \text{as } \kappa \to \infty. \tag{S30}$$

Proof. Because $\mathbf{y}^\star$ is feasible for the hard problem, optimality of $\mathbf{y}^{(\kappa)}$ yields

$$F(\mathbf{y}^{(\kappa)}) + \frac{\kappa}{2}\sum_{i=1}^{r} \rho_i [g_i(\mathbf{y}^{(\kappa)})]_+^2 \le F(\mathbf{y}^\star). \tag{S31}$$

Let $F_{\min} := \min_{\mathbf{y}\in\mathcal{X}} F(\mathbf{y})$ and $\rho_{\min} := \min_i \rho_i > 0$. Compactness of $\mathcal{X}$ and continuity of F ensure that $F_{\min}$ is finite. Equation (S31) implies

$$\sum_{i=1}^{r} [g_i(\mathbf{y}^{(\kappa)})]_+^2 \le \frac{2\big(F(\mathbf{y}^\star) - F_{\min}\big)}{\kappa\rho_{\min}} \longrightarrow 0. \tag{S32}$$

Every accumulation point of $\mathbf{y}^{(\kappa)}$ is therefore hard-feasible. Also, Eq. (S31) gives $F(\mathbf{y}^{(\kappa)}) \le F(\mathbf{y}^\star)$. Hence every accumulation point is a global minimizer of the hard problem. Its uniqueness implies $\mathbf{y}^{(\kappa)} \to \mathbf{y}^\star$.

It remains to prove multiplier convergence. If $i \notin \mathcal{A}_N^\star$, then $g_i(\mathbf{y}^\star) < 0$. Convergence of the primal sequence implies $g_i(\mathbf{y}^{(\kappa)}) < 0$ and consequently $\omega_i^{(\kappa)} = 0$ for all sufficiently large κ.

Let Z have orthonormal columns spanning the tangent subspace of the active box face at $\mathbf{y}^\star$:

$$\operatorname{range}(Z) = \ker((B^\star)^\mathsf{T}). \tag{S33}$$

Any bound inactive at $\mathbf{y}^\star$ remains inactive at $\mathbf{y}^{(\kappa)}$ for sufficiently large κ. On a coordinate active at $\mathbf{y}^\star$, the penalty minimizer is either interior or on the same bound once it is sufficiently close to $\mathbf{y}^\star$. Therefore, for every sufficiently large κ and every $\mathbf{n}^{(\kappa)} \in N_{\mathcal{X}}(\mathbf{y}^{(\kappa)})$ appearing in Eq. (S24),

$$\mathbf{n}^{(\kappa)} \in \operatorname{range}(B^\star), \qquad Z^\mathsf{T}\mathbf{n}^{(\kappa)} = 0. \tag{S34}$$

Set

$$M := Z^\mathsf{T} G_{\mathcal{A}_N^\star}. \tag{S35}$$

The effective-space LICQ makes M full column rank. To see this, if $M\mathbf{c} = 0$, then $G_{\mathcal{A}_N^\star}\mathbf{c} \in \operatorname{range}(B^\star)$; full column rank of $[B^\star \; G_{\mathcal{A}_N^\star}]$ then forces $\mathbf{c} = 0$.

Projecting Eq. (S24) by Z^T and using Eqs. (S34)–(S35) gives

$$Z^\mathsf{T}\nabla F(\mathbf{y}^{(\kappa)}) + M\boldsymbol{\omega}^{(\kappa)}_{\mathcal{A}^\star_N} = 0. \tag{S36}$$

Consequently,

$$\boldsymbol{\omega}^{(\kappa)}_{\mathcal{A}^\star_N} = -(M^\mathsf{T}M)^{-1}M^\mathsf{T}Z^\mathsf{T}\nabla F(\mathbf{y}^{(\kappa)}). \tag{S37}$$

Taking $\kappa \to \infty$ and using $\mathbf{y}^{(\kappa)} \to \mathbf{y}^\star$ gives the unique solution of the projected hard-KKT stationarity equation. By Eq. (S10), that solution is exactly $\boldsymbol{\omega}^{\mathrm{h}}_{\mathcal{A}^\star_N}$. Together with the eventual zeros outside $\mathcal{A}^\star_N$, this proves convergence of the full network multiplier vector. □

S3.3 Iterated smooth-to-hard limit

Theorem S1 (Limiting consistency of the smooth scarcity prices)**.** *Under the assumptions in Section S1,*

$$\lim_{\kappa\to\infty}\left(\lim_{\varepsilon\downarrow 0}\mathbf{y}^{(\kappa,\varepsilon)}\right) = \mathbf{y}^\star, \tag{S38}$$

and

$$\lim_{\kappa\to\infty}\left(\lim_{\varepsilon\downarrow 0}\boldsymbol{\omega}^{(\kappa,\varepsilon)}\right) = \boldsymbol{\omega}^{\mathrm{h}}. \tag{S39}$$

Accordingly,

$$\lim_{\kappa\to\infty}\lim_{\varepsilon\downarrow 0}(A^V)^\mathsf{T}\boldsymbol{\mu}^{(\kappa,\varepsilon)} = (A^V)^\mathsf{T}\boldsymbol{\lambda}^{\mathrm{h}}, \tag{S40}$$

$$\lim_{\kappa\to\infty}\lim_{\varepsilon\downarrow 0}(A^C)^\mathsf{T}\boldsymbol{\nu}^{(\kappa,\varepsilon)} = (A^C)^\mathsf{T}\boldsymbol{\nu}^{\mathrm{h}}. \tag{S41}$$

Proof. Apply Proposition S1 for each fixed κ, and then apply Proposition S2. The price-component limits follow because A^V and A^C are fixed linear maps. □

The two propositions are model-specific versions of standard smoothing and quadratic exterior-penalty results; see Refs. [1, 2, 3] and the variational treatment in Ref. [4].

S4 Evaluation points and finite-parameter interpretation

Theorem S1 evaluates multiplier estimates at the corresponding centralized penalty minimizers. When the limiting multiplier is positive, it is recovered through the $O(1/\kappa)$ positive constraint displacement of the quadratic-penalty solution. For example, consider

$$\min_{y\in[-1,1]} \frac{1}{2}(y-1)^2 \quad \text{subject to} \quad y \le 0. \tag{S42}$$

The hard solution is $y^\star = 0$ with multiplier $\omega^{\mathrm{h}} = 1$, whereas the quadratic-penalty solution satisfies

$$y^{(\kappa)} = \frac{1}{1+\kappa}, \qquad \omega^{(\kappa)} = \frac{\kappa}{1+\kappa} \longrightarrow 1. \tag{S43}$$

Evaluation at the fixed hard solution instead gives $\kappa[y^\star]_+ = 0$, illustrating why the penalty minimizers are the evaluation points in the theorem.

The scaled smoothing bound at a common argument is

$$0 \le \kappa\rho_i s_\varepsilon(z) - \kappa\rho_i[z]_+ \le \kappa\rho_i\varepsilon\log 2. \tag{S44}$$

The stated order, $\varepsilon \downarrow 0$ at fixed κ followed by $\kappa \to \infty$, first removes the smoothing error and then enforces the hard constraints. A joint-limit result would require a suitable scaling condition and its own proof.

At submitted power profiles, including a QNE of the responsibility-aware game, the finite-parameter coefficients provide explicit differentiable scarcity signals. Theorem S1 concerns centralized OPF minimizers, so it does not identify these coefficients as exact hard-OPF dual variables at an arbitrary profile. Numerical Study I quantifies finite-parameter agreement between the centralized softplus and hard-OPF benchmarks.

S5 Source-Attribution Marginal Feedback and Monotonicity

This section concerns the responsibility-settlement VI at fixed finite penalty parameters and $\varepsilon > 0$. Let $\mathbf{p}^0$ be the fixed reference net-power vector, $\Delta\mathbf{p} := \mathbf{p} - \mathbf{p}^0$, and $\mathbf{p}(\tau) := \mathbf{p}^0 + \tau\Delta\mathbf{p}$. The main-study network maps are $\mathbf{v} = v_0\mathbf{1} - A^V\mathbf{p}$ and $\mathbf{f} = A^C\mathbf{p}$; the corresponding violations are $z_m^V = \underline{v} - v_m$ and $z_\ell^C = f_\ell - \overline{f}_\ell$. The row sets $\mathcal{R}_V$ and $\mathcal{R}_C$ contain the constrained buses and monitored lines, respectively. For $X \in \{V, C\}$, the nodal price and its source-attributed change are

$$\pi_k^X(\mathbf{p}) = \sum_{r\in\mathcal{R}_X} A_{rk}^X \kappa_X s_\varepsilon(z_r^X(\mathbf{p})),$$

$$\pi_{m\leftarrow k}^{X,B} = \Delta p_k \int_0^1 \frac{\partial \pi_m^X(\mathbf{p}(\tau))}{\partial p_k}\, d\tau.$$

The feedback added to the core own-decision marginal is

$$\Delta_k^{\text{attr}} = \sum_{X\in\{V,C\}} \left[\Delta p_k \frac{\partial \pi_k^X}{\partial p_k} + \sum_m p_m^0 \frac{\partial \pi_{m\leftarrow k}^{X,B}}{\partial p_k} \right].$$

The full-space VI operator F_p collects the responsibility-bill marginals minus marginal utility, and $F_y(\mathbf{y}) = G_a^\top F_p(\mathbf{p}^{\text{fix}} + G_a\mathbf{y})$ acts on $\mathcal{Y}^{\text{eff}} = \mathcal{X}$. Here $D_{\text{util}} = \operatorname{diag}(\beta_{u,k})$ is the full nodal-space utility curvature; its effective-space restriction is $D_{\text{util},a} = G_a^\top D_{\text{util}} G_a$, as used in Eq. (S4). The symmetric matrix H^L is the loss matrix in the main manuscript, π^E is the nonnegative wholesale energy price, and v_0 is the fixed root squared voltage.

For $X \in \{V, C\}$ and constraint row $r \in \mathcal{R}_X$, set $a_{rk} := A_{rk}^X$ and $c_r := \sum_m a_{rm} p_m^0$. Throughout this section, s_ε is the softplus function. Along the straight path, define

$$\begin{aligned} z_r^0 &:= z_r^X(\mathbf{p}^0), \qquad b_r := \sum_k a_{rk}\Delta p_k, \\ z_r(\tau) &:= z_r^0 + \tau b_r, \qquad z_r^1 := z_r(1), \\ t_r &:= \kappa_X s_\varepsilon'(z_r^1), \quad u_r := \kappa_X s_\varepsilon''(z_r^1). \end{aligned} \tag{S45}$$

The continuous path moments are

$$\begin{aligned} T_r &:= \int_0^1 \kappa_X s_\varepsilon'(z_r(\tau))\, d\tau, \\ M_r &:= \int_0^1 \tau\kappa_X s_\varepsilon''(z_r(\tau))\, d\tau, \\ N_r &:= \int_0^1 \tau^2\kappa_X s_\varepsilon'''(z_r(\tau))\, d\tau. \end{aligned} \tag{S46}$$

Differentiating the source-attributed price path integrals gives the row feedback

$$\Delta_{k,r}^{\text{attr}} = \Delta p_k a_{rk}^2 t_r + c_r\big(a_{rk}T_r + \Delta p_k a_{rk}^2 M_r\big). \tag{S47}$$

Differentiating with respect to p_j yields

$$\begin{aligned} [J_{\Delta,r}]_{kj} = {} & \delta_{kj} a_{rk}^2 t_r + \Delta p_k a_{rk}^2 a_{rj} u_r \\ & + c_r\big[(a_{rk}a_{rj} + \delta_{kj}a_{rk}^2)M_r \\ & \qquad + \Delta p_k a_{rk}^2 a_{rj} N_r\big], \end{aligned} \tag{S48}$$

where δ_{kj} is the Kronecker delta. Summing over constraint rows and adding utility, loss, and direct voltage/congestion price derivatives gives the complete Jacobian. The direct contribution for channel X is $(A^X)^\top \operatorname{diag}(t_r) A^X$, and the loss contribution is $(2\pi^E/v_0)[H^L + \operatorname{diag}(H^L_{kk})]$.

For $b_r \neq 0$, integration by parts gives

$$\begin{aligned} T_r &= \frac{\kappa_X [s_\varepsilon(z_r^1) - s_\varepsilon(z_r^0)]}{b_r}, \\ M_r &= \frac{t_r - T_r}{b_r}, \qquad N_r = \frac{u_r - 2M_r}{b_r}. \end{aligned} \tag{S49}$$

At zero slope, the exact limits are

$$\begin{aligned} T_r &= \kappa_X s'_\varepsilon(z_r^0), \\ M_r &= \tfrac{1}{2}\kappa_X s''_\varepsilon(z_r^0), \\ N_r &= \tfrac{1}{3}\kappa_X s'''_\varepsilon(z_r^0). \end{aligned} \tag{S50}$$

The fixed-Q solver uses the same Gauss–Legendre rule for the path moments, bills, and operator. The curvature diagnostics use the continuous moments in Eqs. (S49) and (S50). High precision controls cancellation, and segmented quadrature provides the independent check described in the computational solution and validation workflow of the main manuscript.

Let $S_p := (\nabla F_p + \nabla F_p^\top)/2$ and define

$$\begin{aligned} S_{\text{loss}} &:= \frac{2\pi^E}{v_0}[H^L + \operatorname{diag}(H^L_{kk})], \\ S_X^{\text{dir}} &:= (A^X)^\top \operatorname{diag}(t_r) A^X, \\ S_y^{\text{smooth}} &:= G_a^\top [D_{\text{util}} + S_{\text{loss}} + S_V^{\text{dir}} + S_C^{\text{dir}}] G_a. \end{aligned}$$

In the effective decision space, the symmetric Jacobian is

$$\begin{aligned} S_y &= G_a^\top S_p G_a = S_y^{\text{smooth}} + S_{\Delta\text{attr},y}, \\ S_{\Delta\text{attr},y} &= G_a^\top \frac{J_\Delta + J_\Delta^\top}{2} G_a, \qquad J_\Delta = \sum_{X,r} J_{\Delta,r}. \end{aligned} \tag{S51}$$

The core contribution contains positive utility curvature and nonnegative loss and direct-penalty curvature. Equation (S48) shows how source deviations, sensitivities, and the first three softplus derivatives determine the remaining symmetric contribution. This contribution can offset core curvature; its eigenvalues, rather than Jacobian asymmetry alone, determine its effect on monotonicity.

A uniform bound $S_y \succeq \sigma I$ with $\sigma > 0$ certifies QNE uniqueness by the main manuscript's QNE uniqueness theorem. To restate its global certificate, let $J_{\Delta,y} := G_a^\top J_\Delta G_a$ and choose verified bounds

$$\begin{aligned} m_{\text{core}}^{\text{LB}} &\le \inf_{\mathbf{y}\in\mathcal{Y}^{\text{eff}}} \lambda_{\min}(S_y^{\text{smooth}}(\mathbf{y})), \\ |J_{\Delta,y}(\mathbf{y})| &\le M_\Delta \quad \text{elementwise on } \mathcal{Y}^{\text{eff}}, \qquad M_\Delta \ge 0, \\ B_{\Delta\text{attr}}^{\text{ub}} &:= \sqrt{\|M_\Delta\|_1 \|M_\Delta\|_\infty}, \\ \sigma_{\text{LB}}^{\text{glob}} &:= m_{\text{core}}^{\text{LB}} - B_{\Delta\text{attr}}^{\text{ub}}. \end{aligned} \tag{S52}$$

The matrix-norm bound and Weyl's inequality give $\lambda_{\min}(S_y(\mathbf{y})) \ge \sigma_{\text{LB}}^{\text{glob}}$ throughout the feasible box, so a positive bound certifies strong monotonicity. Continuous-operator evaluations instead characterize curvature at the tested operating points. For the signed extension, the same row formulas use the signed constraint gradients, followed by the demand/generation action embedding.

References

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